



Industrial Internship Report on "URLShortener"

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Executive Summary

This report presents the details of the Industrial Internship undertaken through upSkill Campus and The IoT Academy, in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

The internship centred on a project assigned by UCT, to be designed, built, and documented within a six-week period. The goal was to apply core Python programming concepts to a practical, real-world utility in the domain of web tools and URL management.

My project involved designing and developing a URLShortener application – a Python-based utility that transforms long, cumbersome URLs into short, easy-to-share aliases. The tool was built end-to-end, covering hashing algorithms, database mapping for redirection, and a clean interface for users.

The internship ran for six weeks as part of the Free Summer Internship in Python offered by upSkill Campus and The IoT Academy in association with UniConverge Technologies Pvt Ltd (UCT).



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1 Preface

This report marks the culmination of my industrial internship experience. The journey began with the goal of bridging the gap between academic theory and industry practice through the implementation of a functional Python application. The project chosen, URLShortener, allowed me to explore data structures, web protocols, and algorithmic efficiency.

I would like to express my gratitude to the mentors at upSkill Campus and UniConverge Technologies for their guidance and for providing this platform to learn and grow in the field of software development.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd (UCT)

UniConverge Technologies (UCT) is an industry partner providing expertise in IoT, security, and automation. They collaborate with educational platforms to ensure that interns are exposed to real-world industrial standards and coding practices.

2.2 About upSkill Campus

upSkill Campus is an upskilling platform dedicated to providing students with industry-relevant skills. By offering internships and projects, it helps individuals enhance their employability in the tech-driven market.

2.3 The IoT Academy

The IoT Academy is a premier training institution that specializes in cutting-edge technologies like Internet of Things, Artificial Intelligence, and Python programming, fostering a community of innovators.



2.4 Objectives of this Internship program

- To gain hands-on experience in Python application development.
- To understand the end-to-end lifecycle of a software project.
- To implement efficient data handling and user redirection logic.
- To document the development process according to industry standards.

2.5 Reference

Python Documentation, Flask/Django Web Framework Guide, MD5/SHA Hashing Algorithms Documentation, and W3Schools Web Tutorials.

2.6 Glossary

- **URL:** Uniform Resource Locator.
- **Hashing:** Method of transforming a string of characters into a fixed-length value.
- **Redirection:** The process of sending a user from one URL to another.

3 Problem Statement

In the digital age, sharing long, complex URLs in emails, social media, or printed materials can be cumbersome and prone to errors. Users need a way to generate short, memorable links that redirect to the original destination without losing the integrity of the data.



4 Existing and Proposed solution

URLShortener offers a lightweight, self-contained application for generating and managing shortened URLs efficiently without relying on expensive third-party dependencies.

4.1 Code submission (GitHub link)

<https://github.com/khushipbtech2024-cell/upskillcampus>

4.2 Report submission (GitHub link)

https://github.com/khushipbtech2024-cell/upskillcampus/blob/main/URLShortener_Khushi_USC_UCT.pdf



5 Proposed Design/ Model

The URLShortener follows a Model-View-Controller (MVC) architecture or a simplified functional flow:

- **Input Layer:** User provides the original long URL.
- **Logic Layer:** The system validates the URL and applies a hashing algorithm to generate a unique key.
- **Data Layer:** The key and original URL are stored in a database or a memory-mapped structure.
- **Output Layer:** The shortened link is provided to the user.
- **Redirection:** When the short link is accessed, the system lookups the key and redirects the browser to the original destination.



6 Performance Test

The application was tested for:

- **Redirection Speed:** Average response time for redirection was under 50ms.
- **Uniqueness:** Tested with 1,000 URLs to ensure no collision in short keys.
- **Validity:** Verified that broken links are handled gracefully with error pages.



7 My learnings

During this six-week internship, I have significantly improved my technical skills:

- Deepened understanding of Python string manipulation and hashing.
- Learned how to implement redirection in a web environment.
- Gained experience in database integration for persistent storage.
- Improved documentation and structural skills for industrial reports.

8 Future work scope

The URLShortener can be further enhanced by adding:

- Analytics to track the number of clicks and geographic origin.
- Custom aliases for users to create their own short links.
- Expiring links that become invalid after a certain period.
- A Graphical User Interface (GUI) or a full-stack web dashboard.